

Chapter 7: Vectors and Vector Arithmetic – Study Summary

Chapter 7 introduces vectors, one of the foundational data structures in linear algebra and machine learning. A vector is an ordered collection of scalar values. In machine learning, vectors are used to represent feature sets, observations, model coefficients, and target variables. The chapter emphasizes that vectors can be viewed as coordinates in an n-dimensional space and are easily represented in Python using NumPy arrays. The chapter demonstrates how vectors are defined in NumPy and then develops the most important vector operations used throughout machine learning and data science.

What is a Vector?

A vector is an ordered collection of scalar values. Vectors may be represented horizontally or vertically. In machine learning they are frequently used to represent observations, weights, and target variables.

Defining Vectors in NumPy

Vectors are represented using one-dimensional NumPy arrays. Example: `array([1,2,3])`.

Vector Addition

Add corresponding elements of two vectors of equal length. The result is another vector of the same length.

Vector Subtraction

Subtract corresponding elements of two vectors element-by-element.

Vector Multiplication

Chapter 7 focuses on element-wise multiplication. Each component is multiplied by the matching component.

Vector Division

Division is also performed element-by-element between matching entries.

Dot Product

The dot product multiplies corresponding elements and sums the results. It produces a scalar rather than a vector. This operation is extremely important because it is the basis of weighted sums, projections, similarity measures, regression models, and neural networks.

Vector–Scalar Multiplication

A scalar multiplies every component of a vector. This changes the vector's magnitude while preserving direction.

Machine Learning Relevance

The dot product is the most important operation in this chapter because future topics such as linear regression, neural networks, projections, and least squares all depend on it.